

Math 521
Week 1 Assignment

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Problem 2. Let $u, v^1, v^2, \dots, v^n \in C^1(\mathbb{R}^n)$. Let $\mathbf{v} = (v^1, \dots, v^n)$. Show: that

$$\operatorname{div}(u\mathbf{v}) = Du \cdot \mathbf{v} + u \operatorname{div} \mathbf{v}.$$

Proof. We have:

$$\begin{aligned} \operatorname{div}(u\mathbf{v}) &= \sum_{i=1}^n \frac{\partial}{\partial x_i} (uv^i) \\ &= \sum_{i=1}^n (u_{x_i} v^i + u v_{x_i}^i) \\ &= \sum_{i=1}^n u_{x_i} v^i + \sum_{i=1}^n u v_{x_i}^i \\ &= Du \cdot \mathbf{v} + u \operatorname{div} \mathbf{v}. \end{aligned}$$

□

Problem 3 (Evans §2.5 Problem 1). Write down an explicit formula for a function u solving the initial-value problem:

$$\begin{cases} u_t + \mathbf{b} \cdot Du + cu = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\} \end{cases}$$

Here $c \in \mathbb{R}$ and $\mathbf{b} \in \mathbb{R}^n$ are constant.

Proof. Define $z(s) = u(x + s\mathbf{b}, t + s)$. Then:

$$\begin{aligned} \dot{z}(s) &= u_t + \mathbf{b} \cdot Du = -cu(u + s\mathbf{b}, t + s) \\ &= -cz(s) \end{aligned}$$

Solving the ordinary differential equation $\dot{z}(s) = -cz(s)$ gives $z(s) = Ke^{-cs}$, where K is some constant. Note that $z(0) = u(x, t) = K$. Likewise, $z(-t) = u(x - t\mathbf{b}, 0) = g(x - t\mathbf{b}) = Ke^{ct}$. Hence $K = g(x - t\mathbf{b})e^{-ct}$, which establishes $u(x, t) = g(x - t\mathbf{b})e^{-ct}$. □

Problem 4. Use the integration by parts formula and/or the Gauss-Green theorem to establish Green's identities. Let U be a bounded open subset of $\mathbb{R}^n$ with a C^1 boundary ∂U . Then:

- (a) $\int_U \Delta u \, dx = \int_{\partial U} \frac{\partial u}{\partial \nu} \, dS,$
(b) $\int_U \nabla v \cdot \nabla u \, dx = - \int_U u \Delta v \, dx + \int_{\partial U} \frac{\partial u}{\partial \nu} \, dS,$
(c) $\int_U u \Delta v - v \Delta u \, dx = \int_{\partial U} u \frac{\partial v}{\partial \nu} - v \frac{\partial u}{\partial \nu} \, dS.$

Proof. (a) We have:

$$\begin{aligned}
\int_U \Delta u \, dx &= \int_U \sum_{i=1}^n u_{x_i x_i} \, dx \\
&= \sum_{i=1}^n \int_U u_{x_i x_i} \, dx \\
&= \sum_{i=1}^n \left(\int_U u_{x_i} \cdot 0 \, dx + \int_{\partial U} u_{x_i} \nu^i \, dS \right) \\
&= \sum_{i=1}^n \int_{\partial U} u_{x_i} \nu^i \, dS \\
&= \int_{\partial U} \sum_{i=1}^n u_{x_i} \nu^i \, dS \\
&= \int_{\partial U} \nabla u \cdot \nu \, dS \\
&= \int_{\partial U} \frac{\partial u}{\partial \nu} \, dS
\end{aligned}$$

□

(b) Similarly:

$$\begin{aligned}
\int_U (\nabla v \cdot \nabla u) \, dx &= \int_U \left(\sum_{i=1}^n v_{x_i} u_{x_i} \right) \, dx \\
&= \sum_{i=1}^n \int_U u_{x_i} v_{x_i} \, dx \\
&= \sum_{i=1}^n \left(- \int_U u v_{x_i x_i} \, dx + \int_{\partial U} u v_{x_i} \nu^i \, dS \right) \\
&= - \int_U u \sum_{i=1}^n v_{x_i x_i} \, dx + \int_{\partial U} u \sum_{i=1}^n v_{x_i} \nu^i \, dS \\
&= - \int_U u \Delta v \, dx + \int_{\partial U} u \nabla v \cdot \nu \, dS \\
&= - \int_U u \Delta v \, dx + \int_{\partial U} u \frac{\partial v}{\partial \nu} \, dS.
\end{aligned}$$

(c)

Problem 5 (Evans §2.5 Problem 2). *Prove that Laplace's equation $\Delta u = 0$ is rotation invariant. Specifically, let $A = (a_{ij})$ denote an $n \times n$ orthogonal matrix. For $x \in \mathbb{R}^n$, let $y = Ax$; that is:*

$$Ax = y = (y_1, \dots, y_n), \quad y_i = \sum_{j=1}^n a_{ij}x_j$$

Assume $u \in C^2(\mathbb{R}^n)$ satisfies $\Delta u = 0$. Then define

$$v(x) = u(Ax) = u(y).$$

Show that $\Delta v = 0$.

Proof. Note that $\frac{\partial y_k}{\partial x_i} = a_{ki}$. Hence:

$$\begin{aligned} v_{x_i}(x) &= \frac{\partial}{\partial x_i} u(y) \\ &= \sum_{k=1}^n u_{y_k}(y) \frac{\partial y_k}{\partial x_i} \\ &= \sum_{k=1}^n u_{y_k}(y) a_{ki}. \end{aligned}$$

Taking a second partial derivative with respect to x_i gives:

$$\begin{aligned}
 v_{x_i x_i}(x) &= \frac{\partial}{\partial x_i} \sum_{k=1}^n u_{x_k}(y) a_{ki} \\
 &= \sum_{k=1}^n \sum_{j=1}^n u_{x_k x_j}(y) \frac{\partial y_j}{\partial x_i} a_{ki} \\
 &= \sum_{j=1}^n \sum_{k=1}^n a_{ji} a_{ki} u_{x_j x_k}.
 \end{aligned}$$

Summing from $i = 1, \dots, n$ finally yields:

$$\begin{aligned}
 \Delta v(x) &= \sum_{i=1}^n v_{x_i x_i}(x) \\
 &= \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n a_{ji} a_{ki} u_{x_j x_k}(y) \\
 &= \sum_{j=1}^n \sum_{k=1}^n \left(\sum_{i=1}^n a_{ji} a_{ki} \right) u_{x_j x_k}(y) \\
 &= \sum_{j=1}^n \sum_{k=1}^n \delta_{jk} u_{x_j x_k}(y) \\
 &= \sum_{k=1}^n u_{x_k x_k} \\
 &= \Delta u(y) \\
 &= 0
 \end{aligned}$$

□